

Multiple regression and omitted-variable bias

Slide route: Lecture5v2.tex: OVB, multiple regression, goodness of fit, multicollinearity, controls and conditional mean independence. Chapter 6.

Session 7 | Comparing like with like

Outcomes: Predict the sign of omitted-variable bias, interpret a partial coefficient, distinguish perfect from high multicollinearity, and choose a reference category.

100-minute route

0–10	Post-midterm conceptual retrieval.
10–30	OVB slides and education/ability board example.
30–45	Multiple regression and holding controls fixed.
45–55	Goodness of fit and pause.
55–75	Multicollinearity and the dummy-variable trap.
75–90	Control variables and conditional mean independence.
90–100	Exit questions and next topic.

Slide-by-slide script

Slide 1 | Title

Guide Open by asking what changes when a regression compares units that differ in several ways.

Slide 2 | Roadmap

Guide Preview OVB, multiple regression, fit, multicollinearity, and controls. Keep the phrase “holding fixed” visible.

Slide 3 | Last lesson

Guide Retrieve the simple-regression slope and ask what makes its causal interpretation fail.

Slide 4 | Last lesson

Guide Use the class-size example to distinguish a partial relationship from an unconditional comparison.

Slide 5 | OVB

Guide Introduce an omitted ability variable. Ask students to predict the sign before seeing the formula.

Slide 6 | OVB

Guide Write the two links required for bias: ability affects wages and ability is correlated with education.

Slide 7 | OVB

Guide Work simple slope = $.06 + .40(.10) = .10$. Circle the bias component .04.

Slide 8 | OVB

Guide Explain that the sign rule depends on both links. If either link is zero, this simple omitted-variable bias disappears.

Slide 9 | OVB

Guide Ask whether adding a control automatically creates a causal estimate. No; the control can be measured poorly or be endogenous.

Slide 10 | OVB

Guide Use a diagram with education, ability, and wage. Have students point to the backdoor path.

Slide 11 | Identifying causal effects

Guide Return to the intervention question: what would happen to one person under a different education level?

Slide 12 | Identifying causal effects

Guide Contrast observed comparisons with comparisons holding relevant determinants fixed.

Slide 13 | Another way to see this

Guide Use residualization language: remove the part of education explained by controls, then compare the remaining education variation.

Slide 14 | Another way to see this

Guide Warn that partialling out is an algebraic equivalence, not a guarantee that the controls are valid.

Slide 15 | What to do about OVB

Guide Ask students to propose randomization, a credible instrument, or a defensible control strategy.

Slide 16 | Multiple regression model

Guide Write $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$ and say exactly what β_1 holds fixed.

Slide 17 | Multiple regression model

Guide Use a two-regressor prediction example and identify which coefficient changes when the second regressor is added.

Slide 18 | Example for $k = 2$

Guide Calculate one fitted value from the displayed coefficients. Ask students to keep the intercept and both regressors.

Slide 19 | Example for $k = 2$

Guide Interpret a coefficient as a one-unit change in one regressor at fixed values of the others.

Slide 20 | OLS estimator in multiple regression

Guide Connect the normal equations to orthogonality with every included regressor.

Slide 21 | OLS estimator in multiple regression

Guide Ask whether residual orthogonality proves exogeneity. No; it is imposed by the minimization.

Slide 22 | Subsidized meals example

Guide Read the outcome, treatment, controls, and sample before interpreting the coefficient.

Slide 23 | Subsidized meals example

Guide Compare a short and long specification. Ask which coefficient changes and what that suggests about confounding.

Slide 24 | Goodness of fit

Guide Show that adding regressors cannot lower in-sample R^2 on the same observations.

Slide 25 | Goodness of fit

Guide Warn that a higher R^2 is not evidence of a better causal model.

Slide 26 | Goodness of fit

Guide Use adjusted R^2 only as a fit summary, not as a substitute for theory or design.

Slide 27 | OLS assumptions for causal inference

Guide List random sampling, variation, conditional mean zero, and finite moments. Ask which assumption is causal.

Slide 28 | Multicollinearity

Guide Distinguish perfect collinearity, which prevents separate estimation, from high collinearity, which inflates uncertainty.

Slide 29 | Perfect multicollinearity

Guide Use $X_3 = X_1 + X_2$ as a quick algebraic example. Ask which coefficient is not separately identified.

Slide 30 | High multicollinearity

Guide Explain that large standard errors do not imply bias. Precision and identification are different.

Slide 31 | Dummy-variable trap

Guide For categories A, B, and C with an intercept, drop one indicator. Ask students to name the reference group.

Slide 32 | Omitted category

Guide Use the fitted means 50, 55, and 47 from this guide. Change the reference category and show that fitted means stay the same.

Slide 33 | Control variables

Guide Ask whether a proposed control is pre-treatment, a confounder, a mediator, or a collider.

Slide 34 | Conditional mean independence

Guide Interpret $\mathbb{E}(u \mid X_1, X_2) = 0$ as no systematic remaining error after conditioning on both regressors.

Slide 35 | Causal assumptions

Guide Return to the wage example and name the assumption that ability is now adequately handled.

Slide 36 | Summary

Guide Have students explain OVB, partial coefficients, and multicollinearity in one sentence each.

Slide 37 | Material

Guide Assign one specification-comparison exercise and ask students to prepare a joint null for the next class.

Worked example | The sign and size of OVB

Use with slides Lecture5v2 slides 5–10: Omitted Variable Bias. Put the education/ability calculation on the board before slide 6 if you want them to predict the sign, then reveal the size calculation as the slide logic develops.

Let the synthetic population model be

$$\log(wage) = \beta_0 + .06 education + .40 ability + u,$$

with $\mathbb{E}(u \mid education, ability) = 0$. Suppose $\text{Cov}(education, ability) / \text{Var}(education) = .10$. Regressing log wage on education alone targets

$$.06 + .40(.10) = .10.$$

The omitted ability term makes the simple-regression slope exceed the partial effect by .04.

Ask before the arithmetic Which two conditions create OVB? The omitted variable must affect the outcome conditional on included regressors, and it must be correlated with the included regressor after accounting for the model's other included controls.

Interpretation Holding ability fixed, the model implies an approximately 6% wage change per additional education year, with an exact model-implied multiplicative change of $e^{.06} - 1 \simeq 6.18\%$. Log interpretation will be developed in session 9.

Side examples | Controls and collinearity

Use with slides Lecture5v2 slides 16–19 for partialling-out intuition, slides 26–32 for multicollinearity and the dummy-variable trap, and slides 33–35 for control variables and conditional mean independence.

Partialling-out intuition First remove the part of education predicted by the controls. Also remove the part of the outcome predicted by the controls. Regress the outcome residual on the education residual. The slope is the education coefficient from the full OLS regression using the same observations. This is a teaching intuition, not an additional assumption.

Dummy trap For three exhaustive categories A, B, C, an intercept and all three indicators satisfy $1 = D_A + D_B + D_C$. Drop one indicator or the intercept. With A as the reference, an intercept of 50 and coefficients 5 on B and -3 on C imply fitted means 50, 55, and 47.

High versus perfect collinearity Perfect collinearity prevents unique identification of all coefficients. High but imperfect collinearity can inflate SEs; it does not by itself bias OLS under the other assumptions.

Goodness of fit Adding a regressor cannot lower in-sample R^2 on the same sample. Adjusted R^2 can fall. Neither chooses valid causal controls automatically.

Teaching comments

Do not say that including every available variable always improves causal inference. A post-treatment mediator can remove part of the effect of interest; a collider can introduce bias. At this level, establish the question and timing of variables before deciding what to control for.

For a treatment effect, conditional mean independence can justify the coefficient on treatment with suitable controls without making every control coefficient a causal effect. Emphasize this distinction in the final control-variable slides.

Exit ticket and answers

- 1 Omitted Z increases Y and is negatively correlated with X . Bias in the simple X coefficient? Negative.
- 2 Does a large SE caused by collinearity prove no relationship? No; precision is weak.
- 3 If the reference category changes, do fitted category means change? No; the parameterization changes.

If short / next preparation Shorten repeated subsidized-meal tables, keeping one comparison of specifications. Read Chapter 7 and prepare to test two coefficients together.

Source: Lecture5v2.tex. Numerical examples are synthetic.